NEXUS-Mars | Technical Specifications

Coordinate systems and scientific requirements • Research record • 24 October 2026 • v0.1

1. Purpose and scope

This specification records the documentary evidence currently established for the Mars Climate Sounder (MCS) source data underlying the NEXUS-Mars Mars Year 29 (MY29) cloud-centroid visualisation. It distinguishes confirmed characteristics of NASA Planetary Data System (PDS) products from unresolved conventions in the downstream Cloudspotting on Mars catalogue.

2. Confirmed scientific requirements

Scientific requirement	Status	Evidence and technical interpretation
Instrument identification	CONFIRMED	Mars Climate Sounder (MCS), carried by Mars Reconnaissance Orbiter (MRO). The PDS catalog identifies the instrument host as MRO and instrument as MCS. [1, 2]
Raw measurement provenance	CONFIRMED	MCS EDR records contain unpacked instrument science and engineering measurements, acquired at approximately 2.048-second intervals. Calibrated RDR products are generated from EDR measurements. [1, 2]
Spectral channel calibration references	CONFIRMED	PDS CALIB documentation identifies MCS_CENTRAL_WAVENUMBERS.ASC and MCS_SPECTRAL_RESPONSE.ASC. The central wavenumber reference documents A1–A5 values and detector-specific B1–B3 values. [3]
Underlying RDR areocentric coordinate convention	CONFIRMED (RDR)	The RDR DATASET.CAT explicitly specifies areocentric spherical coordinates for positions and vectors. Scene and spacecraft geometry is calculated with SPICE/NAIF. This does not independently prove that Cloudspotting MY29 retained every RDR convention unchanged. [1]

2.1 Instrument and processing lineage

The MCS EDR is the unpacked, uncalibrated source record. The MCS RDR contains calibrated radiances and supporting geometry, including scene latitude/longitude calculated using SPICE/NAIF kernels. The Cloudspotting MY29 catalogue is a separate derived citizen-science product; its precise field mappings require validation against its own methodology. [1, 2]

2.2 Spectral calibration reference values

The supplied central-wavenumber document lists: A1 606.916 cm^{-1}; A2 631.017 cm^{-1}; A3 648.703 cm^{-1}; A4 842.724 cm^{-1}; A5 463.436 cm^{-1}. Telescope B channels require detector-specific values. These are calibration references, not geographic coordinates or cloud altitudes. [3]

2.3 Coordinate reference interpretation

Areocentric spherical coordinates describe positions relative to Mars's centre. Accordingly, the underlying RDR convention is compatible with a planetocentric latitude interpretation. The RDR catalog does not, by itself, establish the positive longitude direction or the exact geometric meaning of Scene_lat and Scene_lon in the Cloudspotting centroid CSV. [1]

3. Outstanding validation requirements

Pending: (i) confirm whether MY29 Scene_lat retains the RDR areocentric convention; (ii) determine longitude positive-east versus positive-west; (iii) establish whether Scene_lat/Scene_lon identify tangent-point or other scene geometry; (iv) verify the altitude reference and the relationship between scene geometry and each identified cloud centroid; (v) inspect missing or flagged geometry values where applicable.

4. Implementation boundary

NEXUS-Mars may retain its illustrative unit-sphere representation and plot the supplied scene latitude/longitude as associated observation geometry. Until the pending definitions are resolved, it must not describe plotted points as independently verified three-dimensional cloud positions or imply that the globe models Mars terrain or atmospheric physics.

5. Source register

[1] NASA PDS, MRO Mars Climate Sounder Level 4 RDR V1.0, DATASET.CAT (user-supplied extract); DOI: 10.17189/4w08-hh34. Relevant sections: Processing; RDR Calibration; Ancillary Data; Coordinate System; Quality Flags.

[2] NASA PDS, MRO Mars Climate Sounder Level 2 EDR V1.0, DATASET.CAT (user-supplied extract); DOI: 10.17189/hc7a-2975. Relevant sections: Data Set Overview; Processing; Coordinate System.

[3] NASA PDS, MCS CALIB directory description, MCS_CENTRAL_WAVENUMBERS.ASC and MCS_SPECTRAL_RESPONSE.ASC (user-supplied text, 2007 instrument calibration reference).

Evidence classification: "Confirmed" applies to the identified PDS source products and documents, not automatically to all transformations in the downstream MY29 citizen-science catalogue.